

Capstone Project 1: Online Store Order Analytics

Assignment Tasks

1. Basic Queries

```
db.customers.insertMany([  
  
  { customer_id: 1, name: "Aarav", city: "Hyderabad", membership: "Gold", age: 28 },  
  
  { customer_id: 2, name: "Priya", city: "Bangalore", membership: "Silver", age: 25 },  
  
  { customer_id: 3, name: "Rahul", city: "Mumbai", membership: "Gold", age: 30 },  
  
  { customer_id: 4, name: "Sneha", city: "Delhi", membership: "Silver", age: 27 },  
  
  { customer_id: 5, name: "Kiran", city: "Hyderabad", membership: "Gold", age: 35 },  
  
  { customer_id: 6, name: "Meera", city: "Chennai", membership: "Bronze", age: 22 }  
  
]);
```

```
db.products.insertMany([  
  
  { product_id: 101, name: "Laptop", category: "Electronics", price: 75000, stock: 5 },  
  
  { product_id: 102, name: "Phone", category: "Electronics", price: 50000, stock: 10 },  
  
  { product_id: 103, name: "Desk", category: "Furniture", price: 15000, stock: 7 },  
  
  { product_id: 104, name: "Chair", category: "Furniture", price: 7000, stock: 15 },  
  
  { product_id: 105, name: "Tablet", category: "Electronics", price: 30000, stock: 8 },  
  
]);
```

```
{ product_id: 106, name: "Printer", category: "Electronics", price: 12000, stock: 6 }  
]);
```

```
db.orders.insertMany([  
  { order_id: 1001, customer_id: 1, product_id: 101, quantity: 1, status: "Delivered" },  
  { order_id: 1002, customer_id: 2, product_id: 102, quantity: 2, status: "Pending" },  
  { order_id: 1003, customer_id: 1, product_id: 105, quantity: 1, status: "Cancelled" },  
  { order_id: 1004, customer_id: 3, product_id: 103, quantity: 1, status: "Delivered" },  
  { order_id: 1005, customer_id: 5, product_id: 102, quantity: 3, status: "Delivered" },  
  { order_id: 1006, customer_id: 6, product_id: 104, quantity: 4, status: "Pending" },  
  { order_id: 1007, customer_id: 4, product_id: 106, quantity: 2, status: "Cancelled" },  
  { order_id: 1008, customer_id: 3, product_id: 101, quantity: 1, status: "Delivered" }  
]);
```

```
db.customers.find();
```

```
db.products.find();
```

```
db.orders.find();
```

```
db.customers.find({city:"Hyderabad"});
```

```
db.products.find({category:"Electronics"});
```

```
db.orders.find({status:"Delivered"});
```

2.Filtering Queries

```
db.products.find({price:{$gt:30000}});
```

```
db.products.find({price:{$gte:10000,$lte:50000}});
```

```
db.customers.find({age:{$gt:26}});
```

```
db.orders.find({quantity:{$gt:1}});
```

```
db.products.find({stock:{$lte:10}});
```

```
db.orders.find({status:{$ne:"Cancelled"}});
```

```
db.customers.find({city:{$in:["Hyderabad","Mumbai"]}});
```

3. Projection Queries

```
db.customers.find({}, {_id:0,name:1,city:1});
```

```
db.products.find({}, {_id:0, name:1, category:1, price:1});
```

```
db.orders.find({}, {_id:0, order_id:1, quantity:1, status:1});
```

4. Sorting, Limit, Skip

```
db.products.find().sort({price:1});
```

```
db.products.find().sort({price:-1});
```

```
db.products.find().sort({price:-1}).limit(3);
```

```
db.products.find().sort({price:1}).limit(2);
```

```
db.products.find().sort({price:1}).skip(2);
```

```
db.customers.find().sort({age:-1});
```

5.Update Operations

```
db.products.updateOne({name:"Laptop"},{$set:{price:78000}});
```

```
db.products.updateMany({category:"Electronics"},{$set:{discount:10}});
```

```
db.orders.updateMany({status:"Pending"},{$set:{priority:"High"}});
```

```
db.customers.updateOne({name:"Meera"},{$set:{membership:"Silver"}});
```

6. Delete Operations

```
db.products.deleteOne({name:"Printer"});
```

```
db.products.deleteMany({category:"Furniture"});
```

```
db.orders.deleteMany({status:"Cancelled"});
```

7. Count Queries

```
db.customers.countDocuments();
```

```
db.products.countDocuments({category:"Electronics"});
```

```
db.orders.countDocuments({status:"Delivered"});
```

```
db.customers.countDocuments({city:"Hyderabad"});
```

8. Aggregation Queries

```
db.products.aggregate([  
  $group:{  
    _id:"$category",  
    total_stock:{$sum:"$stock"}  
  }  
]);
```

```
db.products.aggregate([
```

```
$group:{
  _id:"$category",
  avg_price:{$avg:"$price"}
}
});
```

```
db.products.aggregate([{$group:{
  _id:null,
  max_price:{$max:"$price"}
}
});
```

```
db.products.aggregate([{$group:{
  _id:null,
  min_price:{$min:"$price"}
}
});
```

```
db.products.aggregate([{$project:{
  total_value:{$multiply:["$price","$stock"]}
}
});
```

```
db.orders.aggregate([{
  $group:{
    _id:"$product_id",
    total_quantity:{$sum:"$quantity"}
  }
}]);
```

```
db.orders.aggregate([{
  $group:{
    _id:"$customer_id",
    total_quantity:{$sum:"$quantity"}
  }
}]);
```

9. \$lookup Queries

```
db.orders.aggregate([{
  $lookup:{
    from:"customers",
    localField:"customer_id",
    foreignField:"customer_id",
    as:"customer_details"
  }
}]);
```

```
db.orders.aggregate([  
  $lookup:{  
    from:"products",  
    localField:"product_id",  
    foreignField:"product_id",  
    as:"product_details"  
  }  
])
```

```
db.orders.aggregate([  
  $lookup:{  
    from:"customers",  
    localField:"customer_id",  
    foreignField:"customer_id",  
    as:"customer_details"  
  }  
],  
{ $unwind:"$customer_details"},  
{  
  $lookup:{  
    from:"products",  
    localField:"product_id",  
    foreignField:"product_id",  
    as:"product_details"
```

```
    }  
  },  
  {$unwind:"$product_details"},  
  {$project:  
    {  
      _id:0,  
      customer_name:"$customer_details.name",  
      product_name:"$product_details.name"  
    }  
  }  
});
```

```
db.orders.aggregate([  
  $lookup:{  
    from:"customers",  
    localField:"customer_id",  
    foreignField:"customer_id",  
    as:"customer_details"  
  }  
},  
  {$unwind:"$customer_details"},  
  {  
    $lookup:{  
      from:"products",  
      localField:"product_id",
```

```
    foreignField:"product_id",
    as:"product_details"
  }
},
{$unwind:"$product_details"},
{
  $project:{
    _id:0,
    customer_name:"$customer_details.name",
    city:"$customer_details.city",
    product_name:"$product_details.name",
    quantity:1,
    status:1
  }
}
]);
```


